

2 **Goodness-of-fit for multi-distribution neutrino** 3 **cross-section measurements with shared events**

4 **F. Martínez López**

5 *Department of Physics, Indiana University,*
6 *727 E. Third St., Bloomington, IN, USA, 47405*

7 *E-mail:* frmar@iu.edu

8 **ABSTRACT:** When multiple differential cross-section distributions are extracted from a common event
9 sample with correctly evaluated inter-distribution correlations, the statistical covariance matrix is
10 generically singular. Exact linear constraints from event sharing across measurements, such as equal
11 distribution totals when all selected events populate bins in every distribution, produce directions
12 with zero statistical variance. Systematic uncertainties restore formal invertibility but leave near-
13 singular directions that yield unphysically inflated χ^2 values in global goodness-of-fit tests. We
14 present a projection method that restricts the test to the subspace carrying independent statistical
15 information, yielding a χ^2 distributed with $N_{\text{bins}} - N_{\text{null}}$ degrees of freedom, where N_{null} is the
16 number of independent constraints. The method is connected to established results on hypothesis
17 testing with singular covariance matrices and validated with both an analytical toy model and a
18 realistic simulated neutrino–argon cross-section measurement including unfolding and parametrized
19 multi-source systematic uncertainties.

20 **KEYWORDS:** Analysis and statistical methods

21 **ARXIV EPRINT:** [1234.56789](https://arxiv.org/abs/1234.56789)

22	Contents	
23	1 Introduction	2
24	2 Rank deficiency of the statistical covariance	3
25	2.1 Statistical covariance from shared events	3
26	2.2 Combination matrix and null space	4
27	2.3 Types of constraints	5
28	2.4 Effect of systematic uncertainties	6
29	2.5 Ignoring inter-block correlations	6
30	3 The projection method	6
31	3.1 Constructing the projector	6
32	3.2 Projected chi-squared	6
33	3.3 Proof sketch / justification	6
34	3.4 Relation to alternative approaches	6
35	4 Analytical demonstration	6
36	4.1 Setup	6
37	4.2 Eigenvalue analysis	6
38	4.3 Pseudo-experiment validation	6
39	4.4 Sensitivity to threshold choice	6
40	5 Realistic demonstration with neutrino generators	6
41	5.1 Simulation setup	6
42	5.2 Example 1: Two blocks ($N_{\text{null}} = 1$)	6
43	5.3 Example 2: Four blocks with multiplicity slicing ($N_{\text{null}} = 2$)	6
44	5.4 Practical observations	6
45	6 Practical implementation guide	6
46	6.1 Step-by-step recipe	6
47	6.2 What to include in data releases	6
48	6.3 Compatibility notes	6
49	7 Connection to the statistics literature	6
50	8 Summary and recommendations	6

1 Introduction

Modern neutrino cross-section literature is experiencing an increase of published measurements reporting multiple differential distributions extracted from the same event sample. MicroBooNE has reported simultaneous measurements of cross sections for final states with and without protons [1, 2], multi-differential QE-like cross section measurements in kinematic imbalance variables [3, 4], and triple-differential inclusive results [5]. Other relevant examples include the multiple final-state kinematic projections from T2K [6, 7], and double-differential inclusive results from NOvA [8–10]. In each of these analyses, the same selected events populate bins in multiple reported distributions, leading to strong inter-measurement statistical correlations. Separate but related efforts have produced simultaneous measurements across different nuclear targets [11–13] and combined neutrino-antineutrino results [14], where correlations arise from shared systematic uncertainties rather than shared events.

The blockwise unfolding framework [15] provides a rigorous method for computing the full cross-distribution covariance matrices in all such measurements. In this approach, all bins corresponding to the same measured observable in the multi-distribution analysis are organized in blocks, each corresponding to a single differential cross-section distribution. Each bin block is unfolded independently using any standard method (e.g. iterative D’Agostini unfolding [16] or Wiener-SVD [17]), while the full covariance matrix, encoding the correlations between bins from different distributions, is constructed and propagated through the cross section extraction procedure. The statistical covariance for bins in different blocks is determined by the number of events they share. The result is a single combined covariance matrix covering all reported bins across all distributions.

A natural use of these combined measurements is computing global goodness-of-fit metrics, such as a χ^2 statistic, that test a model against all distributions simultaneously. Compared to evaluating each distribution independently, a global test that accounts for all inter-block correlations provides the maximum discriminating power available from the data. Such global metrics are essential for interaction model tuning efforts [18–20], where measurements of multiple kinematic distributions provide complementary sensitivity to different aspects of the underlying physics. However, as we show in this paper, the full inter-block statistical covariance is generically singular when it correctly accounts for correlations from shared events. The rank deficiency arises from exact linear constraints imposed by the event-sharing structure of the binning scheme: certain linear combinations of bins, such as the difference in total event counts between two distributions of the same events, have identically zero variance. While systematic uncertainties formally restore invertibility, the resulting near-singular directions produce unphysically inflated χ^2 values, yielding a global test with the wrong effective number of degrees of freedom.

We present a method based on projecting onto the range (column space) of the statistical covariance matrix that yields a well-defined chi-squared with the correct number of degrees of freedom. We demonstrate the method using a pedagogical toy example and connect it to the established statistical literature on hypothesis testing with singular covariance matrices.

Brief roadmap of sections.

2 Rank deficiency of the statistical covariance

Consider a neutrino cross-section analysis simultaneously measuring B different flux-integrated differential cross sections, each with N_b true bins. If the measurement is extracted from a common event sample then each selected event will populate exactly one bin in each distribution it belongs to. Recent MicroBooNE analyses [21] have adopted the blockwise unfolding approach presented in Ref. [15] to simultaneous cross-section measurements.

This method provides a systematic framework for reporting the correlations between the different extracted differential cross sections. It provides a general prescription to construct statistical covariance matrices that respect the correlations between bins in different distributions or blocks. The full inter-block covariance matrix constructed in reconstructed space is propagated through unfolding using a block-diagonal unfolding matrix, obtained as a direct sum of the individual unfolding matrices computed for each block. The result of the combined measurement is a vector of $N_{\text{bins}} = \sum_{b=1}^B N_b$ cross-section values with an $N_{\text{bins}} \times N_{\text{bins}}$ covariance matrix.

2.1 Statistical covariance from shared events

The core idea behind the treatment of the statistical correlations in the blockwise unfolding approach is the partition of any arbitrary pair of overlapping kinematic bins X and Y into a set of three non-overlapping bins u , v , and w . The events in the intersection $X \cap Y$ are assigned to bin v , while bin u contains the events in the relative complement of Y in X (i.e. $X \setminus Y$), and bin w corresponds to the relative complement of X in Y (i.e. $Y \setminus X$). Denoting the number of predicted events in each bin as n_i , and assuming these follow independent Poisson distributions, the covariance between the counts in bins X and Y is given by [15]:

$$\begin{aligned} \text{Cov}(n_X, n_Y) &= \text{Cov}(n_u + n_v, n_v + n_w) \\ &= \text{Cov}(n_u, n_v) + \text{Cov}(n_u, n_w) + \text{Cov}(n_v, n_v) + \text{Cov}(n_v, n_w) \\ &= \text{Var}(n_v), \end{aligned} \quad (2.1)$$

as by definition bins u , v , and w have no events in common. This implies that the covariance between any two bins can be estimated from the variance of the events they share.

The variance for a set of Monte Carlo (MC) events can be estimated from the sum of their squared weights. This gives us a general prescription to calculate the elements of the statistical covariance matrix as:

$$(C_{\text{stat}})_{ij} = \sum_{e \in (i \cap j)} w_e^2, \quad (2.2)$$

where w_e is the weight associated to event e belonging to the intersection of bins i and j . In the case of measured data the weights are 1, and the covariance element simplifies to the number of events shared between the bins n_{ij} .

After propagation through the block-diagonal unfolding, the statistical covariance matrix in unfolded space inherits the same rank structure, as the unfolding matrices for each individual block are invertible. This rank structure has a direct interpretation in terms of the number of distinct event-sharing patterns across distributions, which we discuss next.

2.2 Combination matrix and null space

The previous discussion shows that the statistical covariance between any two bins is determined solely by the events they share. We now formalize this observation to derive an exact formula for the rank of C_{stat} , and therefore for the number of directions in bin space that carry zero statistical variance.

For each event e , we can define the indicator vector $\phi_e := (\phi_i)_{i=1}^{N_{\text{bins}}}$ such that $\phi_i = 1$ if $i \in \text{bins}(e)$ and $\phi_i = 0$ if $i \notin \text{bins}(e)$, where $\text{bins}(e)$ is the set of bins event e populates across all distributions. The indicator vector acts as an identifier of each event in the combined measurement. For example, in a two-block measurement with 3 bins each, an event populating bin 2 of block 1 and bin 1 of block 2 has indicator vector $\phi_e = (0, 1, 0, 1, 0, 0)$. The statistical covariance matrix can then be re-written in terms of indicator vectors as:

$$C_{\text{stat}} = \sum_e w_e^2 \phi_e \phi_e^\top. \quad (2.3)$$

This outer-product decomposition makes the rank structure of C_{stat} clear. The rank of the statistical covariance matrix equals the dimension of $\text{span}\{\phi_e\}$, which depends only on the distinct indicator vectors present, not on their weights. Since all weights are $w_e > 0$ no indicator vector is removed from the sum, and the column space of C_{stat} is exactly $\text{span}\{\phi_e\}$.

We define the bin-combination matrix A as the matrix whose rows are the unique indicator vectors among $\{\phi_e\}$. Each row of A represent a distinct pattern of bin occupancy across the blocks. By construction the row space of A equals $\text{span}\{\phi_e\}$, and therefore:

$$\text{rank}(C_{\text{stat}}) = \text{rank}(A). \quad (2.4)$$

The number of degenerate directions is then:

$$N_{\text{null}} = N_{\text{bins}} - \text{rank}(A). \quad (2.5)$$

Physically, each of the N_{null} directions corresponds to a linear combination of bins along which the measurement has identically zero variance. Any component of the residual vector along such a direction is not perceived by the statistical covariance; it is a direction that the measurement has no statistical power to constrain. Importantly, both A and N_{null} are directly computable from the MC event-to-bin mapping alone, before constructing any covariance matrix.

A vector $\mathbf{x}$ is in the null space of C_{stat} if and only if, for every event e :

$$\sum_{i \in \text{bins}(e)} x_i = 0, \quad (2.6)$$

or equivalently $A\mathbf{x} = 0$. Each unique indicator vector acts as a linear constraint on the null-space vectors, requiring that the components of $\mathbf{x}$ sum to zero over the bins populated by the different events. The null space directions can be divided into two physically distinct categories: structural constraints that follow from the block topology alone, and kinematic constraints that depend on which regions of phase space are populated.

155 The number of null directions N_{null} depends on the overlap structure between blocks. The constraints
 156 described above hold exactly when every selected event populates at least one bin in every block, as is
 157 the case when underflow and overflow bins are included. Without such bins, events falling outside
 158 the kinematic range of a given block do not contribute to the inter-block sum rules, potentially
 159 reducing N_{null} . In practice, the inclusion of overflow bins is already recommended for proper
 160 unfolding closure [15], and any well-designed blockwise analysis will therefore exhibit the full set
 161 of constraints. The projection method presented here remains valid regardless of the value of N_{null} ,
 162 including the limiting case $N_{\text{null}} = 0$ where C_{stat} is full rank and the projector reduces to the identity.

163	2.4	Effect of systematic uncertainties
164	2.5	Ignoring inter-block correlations
165	3	The projection method
166	3.1	Constructing the projector
167	3.2	Projected chi-squared
168	3.3	Proof sketch / justification
169	3.4	Relation to alternative approaches
170	4	Analytical demonstration
171	4.1	Setup
172	4.2	Eigenvalue analysis
173	4.3	Pseudo-experiment validation
174	4.4	Sensitivity to threshold choice
175	5	Realistic demonstration with neutrino generators
176	5.1	Simulation setup
177	5.2	Example 1: Two blocks ($N_{\text{null}} = 1$)
178	5.3	Example 2: Four blocks with multiplicity slicing ($N_{\text{null}} = 2$)
179	5.4	Practical observations
180	6	Practical implementation guide
181	6.1	Step-by-step recipe
182	6.2	What to include in data releases
183	6.3	Compatibility notes
184	7	Connection to the statistics literature
185	8	Summary and recommendations
186		Acknowledgments
187		References
188	[1]	MICROBooNE collaboration, <i>First Simultaneous Measurement of Differential Muon-Neutrino</i>
189		<i>Charged-Current Cross Sections on Argon for Final States with and Without Protons Using</i>
190		<i>MicroBooNE Data</i> , <i>Phys. Rev. Lett.</i> 133 (2024) 041801 [2402.19281].
191	[2]	MICROBooNE collaboration, <i>Inclusive cross section measurements in final states with and without</i>
192		<i>protons for charged-current ν_{μ}-Ar scattering in MicroBooNE</i> , <i>Phys. Rev. D</i> 110 (2024) 013006
193		[2402.19216].

- [3] MICROBooNE collaboration, *First Double-Differential Measurement of Kinematic Imbalance in Neutrino Interactions with the MicroBooNE Detector*, *Phys. Rev. Lett.* **131** (2023) 101802 [2301.03706].
- [4] MICROBooNE collaboration, *Multidifferential cross section measurements of $\nu\mu$ -argon quasielasticlike reactions with the MicroBooNE detector*, *Phys. Rev. D* **108** (2023) 053002 [2301.03700].
- [5] MICROBooNE collaboration, *Measurement of three-dimensional inclusive muon-neutrino charged-current cross sections on argon with the MicroBooNE detector*, *Phys. Lett. B* **870** (2025) 139939 [2307.06413].
- [6] T2K collaboration, *Characterization of nuclear effects in muon-neutrino scattering on hydrocarbon with a measurement of final-state kinematics and correlations in charged-current pionless interactions at T2K*, *Phys. Rev. D* **98** (2018) 032003 [1802.05078].
- [7] T2K collaboration, *First T2K measurement of transverse kinematic imbalance in the muon-neutrino charged-current single- π^+ production channel containing at least one proton*, *Phys. Rev. D* **103** (2021) 112009 [2102.03346].
- [8] NOvA collaboration, *Measurement of the double-differential muon-neutrino charged-current inclusive cross section in the NOvA near detector*, *Phys. Rev. D* **107** (2023) 052011 [2109.12220].
- [9] NOvA collaboration, *Measurement of $d^2\sigma/d|\vec{q}|dE_{avail}$ in charged current ν_μ -nucleus interactions at $\langle E_\nu \rangle = 1.86$ GeV using the NOvA Near Detector*, *Phys. Rev. D* **111** (2025) 052009 [2410.05526].
- [10] NOvA collaboration, “Measurement of the double-differential cross section of muon-neutrino charged-current interactions with low hadronic energy in the NOvA Near Detector.” 10, 2024.
- [11] T2K collaboration, *Simultaneous measurement of the muon neutrino charged-current cross section on oxygen and carbon without pions in the final state at T2K*, *Phys. Rev. D* **101** (2020) 112004 [2004.05434].
- [12] MINERvA collaboration, *Simultaneous Measurement of ν_μ Quasielasticlike Cross Sections on CH, C, H₂O, Fe, and Pb as a Function of Muon Kinematics at MINERvA*, *Phys. Rev. Lett.* **130** (2023) 161801 [2301.02272].
- [13] MINERvA collaboration, *Simultaneous Measurement of Muon Neutrino $\nu\mu$ Charged-Current Single π^+ Production in CH, C, H₂O, Fe, and Pb Targets in MINERvA*, *Phys. Rev. Lett.* **131** (2023) 011801 [2209.07852].
- [14] T2K collaboration, *First combined measurement of the muon neutrino and antineutrino charged-current cross section without pions in the final state at T2K*, *Phys. Rev. D* **101** (2020) 112001 [2002.09323].
- [15] S. Gardiner, “Mathematical methods for neutrino cross-section extraction.” 2024.
- [16] G. D’Agostini, *A Multidimensional unfolding method based on Bayes’ theorem*, *Nucl. Instrum. Meth. A* **362** (1995) 487.
- [17] W. Tang, X. Li, X. Qian, H. Wei and C. Zhang, *Data Unfolding with Wiener-SVD Method*, *JINST* **12** (2017) P10002 [1705.03568].
- [18] C. Wilkinson et al., *Testing charged current quasi-elastic and multinucleon interaction models in the NEUT neutrino interaction generator with published datasets from the MiniBooNE and MINERvA experiments*, *Phys. Rev. D* **93** (2016) 072010 [1601.05592].

- 235 [19] MINERvA collaboration, *Tuning the GENIE Pion Production Model with MINERvA Data*, *Phys. Rev.*
236 *D* **100** (2019) 072005 [[1903.01558](#)].
- 237 [20] GENIE collaboration, *Neutrino-nucleus $CC0\pi$ cross-section tuning in GENIE v3*, *Phys. Rev. D* **106**
238 (2022) 112001 [[2206.11050](#)].
- 239 [21] MICROBooNE collaboration, *Measurement of single- and double-differential cross sections for*
240 *mesonless charged-current muon neutrino interactions on argon with final-state protons using the*
241 *MicroBooNE detector*, *Phys. Rev. D* **112** (2025) 112004 [[2403.19574](#)].